

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Statement of the main problem

- We are now interested in solving the following coupled ode system

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases} \quad (1)$$

- So we want our drivers to solve the following general case as efficiently as possible

$$i \partial_t \phi_\omega(x, t) = H(t) \phi_\omega(x, t) + S(t), \quad H(t) = H_0 + F(t)x, \quad (2)$$

- Considering, the solution to the homogeneous problem, without the source

$$i \partial_t U(t, t_0) = (H_0 + F(t)x) U(t, t_0), \quad U(t, t) = \mathbb{1}, \quad (3)$$

we can then, from the Duhamel principle, write

$$\phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t_0) - i \int_{t_0}^t U(t - \tau) S(\tau) d\tau. \quad (4)$$

The above is the exact formulation of our problem.

Geometric propagators

Norm-preserving propagation schemes

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U(t - \tau)S(\tau)d\tau. \quad (4)$$

- Our test system will be a forced QHO, $F(t) = F_0 \cos(\omega t)$ with source

$$\begin{cases} i\partial_t \psi(x, t) &= \left(-(1/2)\partial_x^2 + (1/2)x^2 + F(t)x \right) \psi(x, t), \\ i\partial_t \phi_\omega(x, t) &= \left(-(1/2)\partial_x^2 + (1/2)x^2 + F(t)x \right) \phi_\omega(x, t) - ie^{i\omega t} \partial_x \psi(x, t), \end{cases} \quad (F_0 = \hbar = \omega = m = 1) \quad (5)$$

- The evaluation of both the terms shall be done with matching order methods
- We will try the following geometric (norm-preserving for the homogeneous part) propagators
 - Midpoint Duhamel (order 2)
 - Yoshida symmetric decomposition (order 4)
 - Runge-Kutta 4 (EFRK4)?
 - etc...

Geometric propagators

Convergence criteria

$$\partial_t z = Az + S(y, t). \quad (6)$$

- Assessment of these schemes should be done based on two different tests with the number of steps N , and the time step h are such as

$$N_\ell = N_0 2^\ell, \quad h_\ell = \frac{T}{N_0 2^\ell}. \quad (7)$$

- Absolute convergence

$$p_\ell^{\text{abs}} = \frac{\log(E_{\ell-1}^{\text{abs}}/E_\ell^{\text{abs}})}{\log(2)}, \quad \text{with} \quad E_\ell^{\text{abs}} = |z_\ell(T) - z_{\text{exact}}(T)|. \quad (8)$$

- Self-consistency

$$p_\ell^{\text{self}} = \frac{\log(D_{\ell-1}/D_\ell)}{\log(2)} \quad \text{with} \quad D_\ell = |z_\ell(T) - z_{\ell-1}(T)|, \quad D_\ell \sim Ch_\ell^p. \quad (9)$$

- However if we have to impose the test case to the inhomogeneous Schrödinger equation, then we can realistically only carry the second one

Geometric propagators

Midpoint Duhamel - 2nd order approximation

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U(t - \tau) S(\tau) d\tau. \quad (4)$$

- We can show that the above can be expressed as

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t U(\Delta t/2) S_{n+1/2}, \quad (10)$$

where the source is evaluated at $t + \Delta t/2$, i.e

$$S_{n+1/2} = -ie^{i\omega(t+\Delta t/2)} \partial_x \psi_{n+1/2}, \quad \text{with} \quad \psi_{n+1/2} = U(\Delta t/2)\psi_n. \quad (11)$$

This is the midpoint Duhamel formula, of order 2.

- The driver shall be regarded as decent if the expected order is met

Geometric propagators

Midpoint Duhamel - Results

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t U(\Delta t/2) S_{n+1/2}. \tag{10}$$

$$S_{n+1/2} = e^{i\omega(t+\Delta t/2)} \partial_x \psi_{n+1/2}, \quad \text{with} \quad \psi_{n+1/2} = U(\Delta t/2)\psi_n. \tag{11}$$

t	$\Re \psi(1)$	$\Im \psi(1)$	$\ \psi\ $	$\Re \phi(1)$	$\Im \phi(1)$	$\ \phi\ $
-1.0	1.0000	-0.0000	1.0000	0.0574	-0.0000	0.6248
-0.9	0.9979	-0.0482	1.0000	0.0640	-0.0018	0.6432
-0.5	0.9377	-0.2263	1.0000	0.1193	-0.0066	0.7814
-0.1	0.7927	-0.3337	1.0000	0.1885	-0.0195	0.8795
0.0	0.7484	-0.3437	1.0000	0.2036	-0.0192	0.8823
0.5	0.5428	-0.2995	1.0000	0.2925	0.0198	0.8257
0.9	0.4566	-0.2195	1.0000	0.3965	0.0396	0.8201
1.0	0.4474	-0.2054	1.0000	0.4222	0.0347	0.8347

Ne = 2
Np = 6
-xmin = xrange/2 = 7
-t0 = duration/2 = 1

Refinement level k	Time step Δt	$ \psi^{(k)} - \psi^{(k+1)} $	$ \phi^{(k)} - \phi^{(k+1)} $	Observed order
1	.100	3.6596×10^{-6}	2.7621×10^{-5}	—
2	.050	9.1488×10^{-7}	4.9953×10^{-6}	—
3	.025	2.2872×10^{-7}	1.2522×10^{-6}	$p_\psi = 2.0000$ $p_\phi = 1.9961$

Initial conditions
 $\psi_0(x) = (\pi)^{-1/4} e^{-(x-x_c)^2/2}$
 $\phi_0^\omega(x) = \partial_x \psi_0(x)$
 $x_c = 0$

- Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

Geometric propagators

Yoshida (1990) - 4th-order symmetric decomposition

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U(t - \tau)S(\tau)d\tau. \quad (4)$$

- From the following Strang operator $U(t) = e^{-iE_n h/2} e^{-iV_{\text{ext}}(t)h} e^{-iE_n h/2}$, we compose

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad \text{with} \quad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \quad (12)$$

- More precisely, we can observe the following

Theorem 8.6 (Hairer, Solving ODE II p.554)

If Ψ_h represent a one-step method that is symmetric, of order $p = 2k$ and defined in the domain, then the relations

$$2c_1 + c_2 = 1, \quad 2c_1^{p+1} + c_2^{p+1} = 0, \quad (13)$$

imply that the composition (12) is of order $p + 2$.

If you start from what is commonly known as the split-operator method, then in the above $k = 1$.

Geometric propagators

Yoshida (1990) - Results 1 - Self-consistency test

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t U(\Delta t/2) S_{n+1/2}. \tag{10}$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad \text{with} \quad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \tag{12}$$

t	$\Re \psi$	$\Im \psi$	$\ \psi\ $	$\Re \phi$	$\Im \phi$	$\ \phi\ $
-1.0000	1.000000	0.000000	1.000000	0.0573947	0.0000000	0.624813
-0.5000	0.937711	-0.226288	1.000000	0.119266	-0.0066368	0.781378
-0.3750	0.900255	-0.270304	1.000000	0.142128	-0.0109301	0.822875
-0.1250	0.803565	-0.330083	1.000000	0.184650	-0.0191649	0.877411
0.0000	0.748376	-0.343661	1.000000	0.203625	-0.0191819	0.882245
0.1250	0.692051	-0.346136	1.000000	0.222276	-0.0147149	0.876229
0.3750	0.586884	-0.322034	1.000000	0.265508	0.0066504	0.842303
0.5000	0.542795	-0.299476	1.000000	0.292510	0.0197556	0.825727
1.0000	0.447387	-0.205441	1.000000	0.422166	0.0346930	0.834686

Level	$\Delta t_0 = 0.1$ (relative)	E_ψ	E_ϕ
1	Δt_0	4.22×10^{-6}	1.45×10^{-3}
2	$\Delta t_0/2$	2.44×10^{-7}	7.93×10^{-4}
3	$\Delta t_0/4$	1.50×10^{-8}	1.97×10^{-4}
Observed order		$p_\psi = 4.03$	$p_\phi = 2.01$

■ Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

Ne = 2
Np = 6
-xmin = xrange/2 = 7
-t0 = duration/2 = 1

Initial conditions
 $\psi_0(x) = (\pi)^{-1/4} e^{-(x-x_c)^2/2}$
 $\phi_0^\omega(x) = \partial_x \psi_0(x)$
 $x_c = 0$

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

Geometric propagators

Yoshida (1990) - Results 2 - Convergence against a reference solution

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t U(\Delta t/2) S_{n+1/2}. \tag{10}$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad \text{with} \quad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \tag{12}$$

t	$\Re \psi$	$\Im \psi$	$\ \psi\ $	$\Re \phi$	$\Im \phi$	$\ \phi\ $
-1.0000	1.000000	0.000000	1.000000	0.0573947	0.0000000	0.624813
-0.5000	0.937711	-0.226288	1.000000	0.119266	-0.0066368	0.781378
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0.5000	0.542795	-0.299476	1.000000	0.292510	0.0197556	0.825727
1.0000	0.447387	-0.205441	1.000000	0.422166	0.0346930	0.834686

Δt	E_ψ	E_ϕ	P_ψ	P_ϕ
1.0×10^{-1}	7.79×10^{-7}	2.28×10^{-4}	-	-
5.0×10^{-2}	4.85×10^{-8}	2.82×10^{-5}	4.01	3.01
2.5×10^{-2}	3.01×10^{-9}	3.44×10^{-6}	4.01	3.04
Reference ($\Delta t = \Delta t_0/2^4$)	-	-	-	-

Ne = 2
Np = 6
-xmin = xrange/2 = 7
-t0 = duration/2 = 1

Initial conditions

$\psi_0(x) = (\pi)^{-1/4} e^{-(x-x_c)^2/2}$
 $\phi_0^\omega(x) = \partial_x \psi_0(x)$
 $x_c = 0$

■ Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$